

DS PREVIOUS FINAL QUESTION .

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Department of Computer Science and Engineering

B. Sc. in CSE Semester Final Examination, Spring 2024

Course Code: CSE 2321 Course Title: Data Structures

Time: 2 hours 30 minutes

Total marks: 50

[Answer the following questions. Figures in the right hand margin indicate full marks.]

CO DL

Group A

1. a) Define Ackermann function. Find the value of $A(1,4)$ using the definition of Ackermann function. 3 CO4 C5
 1. b) Write the overflow conditions of array representation of circular queue. Explain the reason of the conditions. 3 CO4 C5
 1. c) Explain the Tower of Hanoi problem. Also analysis the complexity of Tower of Hanoi problem. 4 CO4 C5
- OR
1. c) Explain the two dimensional array representation of priority queue. Explain the advantages and disadvantages of priority queue over the general queue. 4

3. a) What is *stack*? Write a procedure to **PUSH** an item onto a stack and **POP** an item from the stack.
- b) Suppose the following stack of integers is in memory where **STACK** is allocated $N = 8$ memory cells:
TOP = 4 STACK: 11, 22 33, 44, __, __, __, __
Describe the **STACK** and **TOP** as the following operations take place—
 - i) **POP (STACK, ITEM)** iii) **PUSH (STACK, 99)**
 - ii) **PUSH (STACK, 77)** iv) **POP (STACK, ITEM)**
- c) Consider the following postfix expression,
P: 3, 2, +, 2, ^, 7, 4, -, 2, *, +, 5, -
Evaluate P, using the algorithm you have studied.
- d) Consider the following *infix* expression Q:
Q: ((A + B) / D) ^ ((E - F) * G)
Translate Q into its equivalent *postfix* expression P using the algorithm you studied.
Or
Write an algorithm that receives a bracket sequence and tells whether the sequence is correct or incorrect.
For example, "()", "()[]{}", "({ })[]" are correct sequences while, "({ })[])", "{ }" are not.
[You can safely assume that the required data structure and its operational algorithms are available to you to solve this problem. You don't need to re-implement them.]

Spring_24

Group B

3. a) Suppose the following characters are stored in an array A:
D, A, T, A, S, T, R, U, C, T, U, R, E, S
 Apply *selection sort* algorithm to sort the array A and show each pass separately. 3 CO3 C2

OR

Suppose the following numbers are stored in an array A:

44, 33, 11, 50, 90, 70, 45, 98, 22, 65, 78

Apply *merge sort* algorithm to sort the array A and show each pass separately.

- b) Consider a situation where *swap* operation is very costly. Which of the following sorting algorithms should be preferred so that the number of swap operations are minimized in general? 1 CO4 C4
 Why?

i) Heap Sort ii) Selection Sort ii) Insertion Sort iv) Merge Sort

- c) Write an algorithm to sort an array A of n elements using *insertion sort*. What is the time complexity of this algorithm? 3 CO1 C2

- d) The following values are to be stored in a hash table:

35, 58, 102, 79, 131, 46, 112, 177, 2XX

3 CO1 C2

Describe how the values are hashed by using *division method* of hashing with a *table size* of 11.
 Use *linear probing* as the method of *collision resolution*.

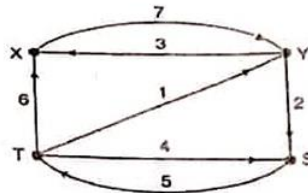
[Here XX is the last two digits of your ID. For example, if ID is C191085, 2XX will be 285].

4. a) Analyze the differences between complete binary tree and heap tree with figure. Build a max heap from the following list of numbers: 5 CO5 C2
33, 29, 49, 21, 57, 62, 73, 54
 b) Write the difference between binary search tree and m-way search tree. How do you delete a node from the binary search tree? Explain the deletion with example. 5

OR

- c) Suppose the following sequences list the nodes of a binary tree T in preorder and inorder, respectively: 5 CO3 C2
 Preorder: G, B, Q, A, C, K, F, P, D, E, R, H
 Inorder: Q, B, K, C, F, A, G, P, E, D, H, R
 Draw the tree.

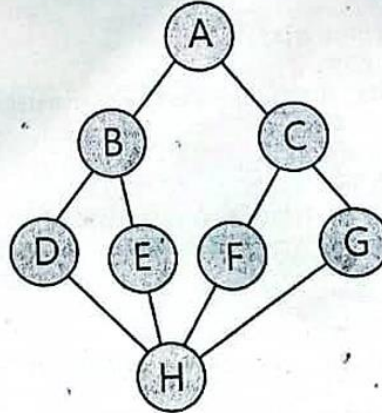
5. a) Use *Warshall's algorithm* to find the *path matrix* for the graph G in following figure. Show each step. 5 CO1 C2



- b) Define *graph*? Draw a picture of the directed graph specified below: 5 CO4 C2
 $G = (V, E)$
 $V(G) = \{1, 2, 3, 4, 5, 6\}$
 $E(G) = \{(1, 2), (2, 3), (3, 4), (5, 1), (5, 6), (2, 6), (1, 6), (4, 6), (2, 4)\}$

Obtain the following for the above graph:

- Find the *adjacency matrix* A of the graph G.
 - Find the *adjacency list* of the graph G.
- OR
- b) Traverse the graph G shown below in **breadth first order**, **depth first order** and construct the **breadth first** and **depth first spanning trees**. Start from node D [if last digit of your ID is *odd*] / G [if last digit of your ID is *even*]. 5



International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Semester Final Examination, Autumn 2023

Course Code: CSE 2321 Course Title: Data Structures

Time: 2 hours 30 minutes

Total marks: 50

[Answer the following questions. Figures in the right hand margin indicate full marks.]

CO DL

Group A

[Answer the following questions]

- a) Write the two properties of a recursion function. Find the base criteria of the following recursive functions. 3 CO4 C5

$$i. H(N) = \begin{cases} 3 * N & \text{IF } N < 5 \\ 2 * H(N - 5) + 7 & \text{Otherwise} \end{cases}$$

$$ii. F(N) = \begin{cases} 1 & \text{IF } N = 0 \\ N.F(N - 1) & \text{IF } N > 0 \end{cases}$$

$$iii. GCD(A, B) = \begin{cases} GCD(B, A) & \text{IF } A < B \\ A & \text{IF } B = 0 \\ GCD(B, MOD(A, B)) & \text{Otherwise} \end{cases}$$

- b) What do you mean by Tower of Hanoi problem? Draw a schematic diagram to solve the Tower of Hanoi problem for n=4 disk. 5 CO4 C5

OR

- b) Find the value of factorial five (5!) using recursion. Show each step and also show the status of stack used in the recursion. 5 CO4 C5
- c) The value of REAR is increased by 1 in case of insertion in queue and the value of FRONT is increased by 1 in case if deletion from queue. Do you agree with this statement and why? 2 CO4 C5

- 3.a) Suppose STACK is allocated $N = 8$ memory cells and initially STACK is empty (i.e. $TOP = 0$). Find the output of the following module [show the elements of STACK and value of TOP in each step]- 2 CO1

1. Set $A := 15$ and $B := 17$
2. Call PUSH (STACK, A)
Call PUSH (STACK, 5)
Call PUSH (STACK, $B - 3$)
Call PUSH (STACK, 10)
Call PUSH (STACK, $A * B$)
Call PUSH (STACK, 9)
3. Repeat while $TOP \neq 0$:
Call POP (STACK, ITEM)
Write: ITEM
[End of loop]
4. Return

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- b) Write an algorithm/code to *reverse* the elements of a stack. 2.5 CO3

- c) Given a postfix expression: 2.5 CO3

Q: 9, -4, +, 3, ↑, N, 4, -, 2, *, +, 5, -

For what value of N, the result of the expression Q would be XY.

[Here, XY is the last two digits of your ID. For example, if ID = C161026, then XY = 26. The result of any division operator must be taken 2 places after the decimal point. For example, $26 / 5 = 5.20$]

- d) Consider the following *infix* expression Q: 3 CO3

Q: $A * (B - D) \uparrow E + F * (G - H / K)$

Translate Q into its equivalent *postfix* expression P using the algorithm you studied.

OR

Suppose a stack supports the following operations in constant time complexity ($O(1)$):

push(x): Pushes an element x onto the stack.

pop(): Removes the element on top of the stack.

top(): Retrieves the element on top of the stack.

getMin(): Retrieves the minimum element in the stack.

Now write algorithms/code of the push(x) and getMin() functions for this special stack

Fig-2c

- d) A DEQUE is a data structure that allows insertion and deletion of elements from both ends that allows the following four operations on it: 2 CO1 C3

1. PUSH_FRONT(X): This function adds an item X in front of its elements.
2. PUSH_BACK(X): This function adds an item X at the end of its elements.
3. POP_FRONT(): This function deletes the first element (if there is any).
4. POP_BACK(): This function deletes the last element (if there is any).

Suppose you are building a DEQUE where the underlying data structure would be a one way linked list. What would be the *time complexity* of these four operations? Explain.

Group B
[Answer the following questions]

3. a) What do you mean by the complexity of algorithms? Analyse the complexity of selection sort algorithm. 5 C
- b) Suppose the following characters are stored in an array A:
M, U, S, E, U, M
Apply *selection sort* algorithm to sort the array A and show each pass separately. 5 C
4. a) Analyse the differences between Binary tree and Complete binary tree with figure. Construct an *expression tree T* for the following arithmetic expression:
 $(A + B * C) + ((D / E - F) * G)$
Traverse the tree T in **preorder**, **postorder** and **inorder**. 5 C
- b) Suppose the following eight numbers are inserted in order into an empty binary search tree T:
50, 23, 44, 22, 77, 35, 60, 40
Draw the tree T and show each step. 4 C
- OR**
- b) What do you mean by *max heap*? Build a max heap from the following list of numbers:
23, 29, 49, 21, 57, 62, 73, 54 4 C
- c) Explain which data structure is most efficient to find the *top 10 largest items* out of *1 million items* stored in file? Why? 1 C
- i) Min heap ii) Max heap iii) BST iv) Sorted array

5. a) What is a *graph*? Consider the following graph G in Fig-5a: 2.5 C

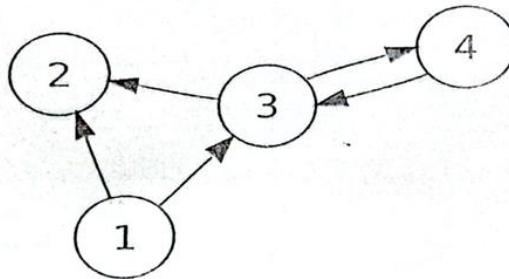
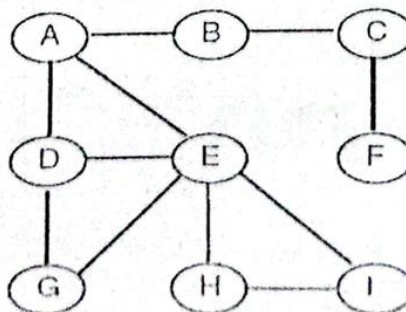


Fig-5a

- i) Find the *adjacency matrix A* of the graph G.
ii) Find the *adjacency list* of the graph G.
- b) Use *Warshall's algorithm* to find the *path matrix* for the graph G in Fig 5a. Show each step. 3.5 C
- OR**
- b) Find the *path matrix* for the graph G in Fig 5a using powers of the adjacency matrix A. 3.5 C
- c) Traverse the graph G shown below in **breadth first order**, **depth first order** and construct the **breadth first** and **depth first spanning trees**. Start from node A [if last digit of your ID is odd] / E [if last digit of your ID is even]. 4 C



[Answer all the following questions. Figures in the right hand margin indicate full marks. Use a
Separate answer script for Group-A and Group-B.]

Group A

- CO DL
1. What is *circular queue*? Write a procedure to *insert* an element into a circular queue and *delete* an element from a circular queue. 4 CO1 C1

OR

What is *priority queue*? Discuss the *one way list* representation of a priority queue. Compare the *one way list* and *array* representation of priority queues.

- Consider the following queue of characters where QUEUE is a circular array which is allocated six memory cells: 2 CO1 C2
- b) FRONT = 2, REAR = 5, QUEUE: _, D, C, R, K, _
Describe the queue, including FRONT and REAR, as the following operations take place –
i) S is added to the queue. iii) B, R, D are added to the queue.
ii) Two letters are deleted. iv) Three letters are deleted.
- c) Consider the code fragment written in C below: 2 CO1 C2

```
void rec(int n)
{
    if (n == 0)
        return;

    printf("%d", n%2);
    rec(n/2);
}
```

What does rec (2X) print? Explain.

[Here X is the last digit of your ID. For example, if ID is C191085, 2X will be 25].

- d) Let A and B be non-negative integers. Suppose a function GCD is recursively defined as follows- 2 CO1 C2

$$\text{GCD}(A, B) = \begin{cases} \text{GCD}(B, A) & \text{if } A < B \\ A & \text{if } B = 0 \\ \text{GCD}(B, \text{MOD}(A, B)) & \text{Otherwise} \end{cases}$$

Here MOD (A, B) denotes the remainder when A is divided by B.

- i) Find GCD (6, 15), GCD (20, 28) and GCD (540, 279).
ii) What does this function do?

OR

Let B be an integer array with N elements. Suppose Y is an integer function defined by

$$Y(K) = Y(B, N, K) = \begin{cases} 0 & \text{IF } K = 0 \\ Y(K-1) + B(K) & \text{IF } 0 < K \leq N \\ Y(K-1) & \text{IF } K > N \end{cases}$$

Find Y(5) for each of the following array:

- i) N = 8, B: 3, 7, -2, 5, X, -4, 2, 7
ii) N = 3, B: X, 7, -4

[Here X is the last digit of your ID. For example, if ID is C191085, X will be 5].

3. a) Write a procedure to PUSH an item onto a stack and POP an item from the stack.
- b) Suppose the following stack of integers is in memory where STACK is allocated N = 8 memory cells:
 TOP = 4 STACK: 11, 22 33, 44, __, __, __, __
 Describe the stack and TOP as the following operations take place –
 i) POP (STACK, ITEM) iii) PUSH (STACK, 66)
 ii) PUSH (STACK, 55) iv) POP (STACK, ITEM)
- c) Consider the following postfix expression,
P: 3, 1, -, 2, ↑, N, 4, /, 2, *, +, 5, -
 For which value of N, the result of this expression would be XY, where XY is the last 2 digits of your ID? [If your ID is C161026, then XY = 26] Use the algorithm you studied to solve this problem.
- d) Consider the following *infix* expression Q:
 Q: $(A * B \uparrow D) / (E - F) + G$
 Translate Q into its equivalent *postfix* expression P using the algorithm you studied.
- Or
 Write an algorithm that receives a bracket sequence and tells whether the sequence is correct or incorrect.
 For example, "()", "()[]{ }", "({ })[]" are correct sequences while, "({ })[])", "{ }" are not.
 [You can safely assume that the required data structure and its operational algorithms are available to you to solve this problem. You don't need to re-implement them.]

sp_23

Group B

3. Suppose the following characters are stored in an array A: 3 CO3 C2
 a) D, A, T, A, S, T, R, U, C, T, U, R, E, S
 Apply *insertion sort* algorithm to sort the array A and show each pass separately.
- OR
 Suppose the following numbers are stored in an array A:
 32, 22, 49, 68, 92, 45, 54, 105, 12, 88, 64, 39
 Apply *merge sort* algorithm to sort the array A and show each pass separately.
- Explain which of the following is not a *stable* sorting algorithm in its typical implementation- 1 CO4 C4
 b) i) Insertion Sort ii) Merge Sort iii) Quick Sort iv) Bubble Sort
- c) Write an algorithm to sort an array A of n elements using *selection sort*. 3 CO1 C2
- d) What is *hashing*? Briefly discuss about any two techniques for *collision resolution* in hashing. 3 CO1 C2
4. Construct an *expression tree* T for the arithmetic expression: CO1 C2
 a) $(A + B * C) - ((D / E + F) - G)$ 3
 Traverse the tree T in *pre*, *post* and *inorder*.

- b) A binary tree T has 9 nodes. The *inorder* and *preorder* traversals of T yields the following sequences of nodes: 3 CO1 C

Inorder : E A C K F H D B G

Preorder : F A E K C D H G B

Draw the tree T.

- c) What is a *binary search tree*? Suppose the following characters are inserted in order into an empty binary search tree. 4 CO1 C

B, A, N, G, L, A, D, E, S, H

i) Draw the tree T.

ii) Find the *inorder* traversal of T.

OR

Illustrate the operation of *Heap sort* on the array $A = (5, 13, 2, 25, N, 7, 17, 20, 8, 4)$. (Note: N is the sum of last two digits of your ID. i.e. if your ID is C151216, then $N = 1 + 6 = 7$)

5. Describe the *adjacency list* and *linked* representation of graph with necessary figure. 5 CO3 C

- a) Define *graph*? Draw a picture of the directed graph specified below: 5 CO4 C

$G = (V, E)$

$V(G) = \{1, 2, 3, 4, 5, 6\}$

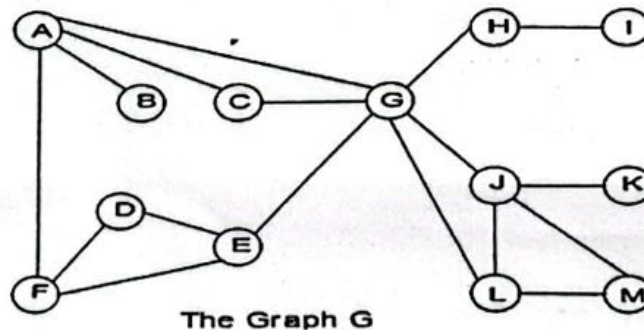
$E(G) = \{(1, 2), (2, 3), (3, 4), (5, 1), (5, 6), (2, 6), (1, 6), (4, 6), (2, 4)\}$

Obtain the following for the above graph:

- Find the *adjacency matrix* A of the graph G.
- Find the *adjacency list* of the graph G.

OR

· Traverse the graph G shown below in *breadth first search* and *depth first search* starting from node C. 5 CO4 C



[Answer *all* the following questions. Figures in the right hand margin indicate full marks. Use a Separate answer script for Group-A and Group-B.]

Group A

- | | CO | DL |
|--|----|--------|
| 1. a) Let B be an integer array with N elements. Suppose Y is an integer function defined by | 2 | CO4 C5 |

$$Y(K) = Y(B, N, K) = \begin{cases} 0 & \text{IF } K = 0 \\ Y(K-1) + B(K) & \text{IF } 0 < K \leq N \\ Y(K-1) & \text{IF } K > N \end{cases}$$

Find Y(5) for each of the following array:

- i. N=8, B: 3, 7, -2, 5, 6, -4, 2, 7 ii. N=3, B: 2, 7, -4

- | | | |
|---|---|--------|
| b) Find the value of A(2,2) using the definition of Ackermann function. | 5 | CO4 C5 |
|---|---|--------|

OR

- | | | |
|---|---|--------|
| b) Find the value of factorial five (5!) using recursion. Show each step and also show the status of stack that is used in recursion. | 5 | CO4 C5 |
|---|---|--------|

- | | | |
|--|---|--------|
| c) Suppose a queue is maintained by a circular array QUEUE with N=12 memory cells. Find the number of elements in QUEUE if | 3 | CO4 C5 |
|--|---|--------|

- i) FRONT = 5, REAR = 10
- ii) FRONT = 12, REAR = 3 and
- iii) FRONT = 6, REAR = 6

- | | | |
|---|-----|--------|
| 3. a) What is <i>stack</i> ? Write a procedure to PUSH an item onto a stack and POP an item from the stack. | 2.5 | CO1 C2 |
|---|-----|--------|

OR

If P be an arithmetic expression written in *postfix expression*, write an algorithm to evaluate P, using a stack to hold operands

- | | | |
|--|---|--------|
| b) Suppose STACK is allocated N= 6 memory cells and initially STACK is empty (i.e. TOP=0). Find the output of the following module [show the elements of STACK and value of TOP in each step]- | 2 | CO1 C2 |
|--|---|--------|

1. Set AAA := 10 and BBB := 20
2. CALL PUSH(STACK, AAA)
CALL PUSH(STACK, 5)
CALL PUSH(STACK, BBB+3)
CALL PUSH(STACK, 15)
CALL PUSH(STACK, AAA+BBB)
3. Repeat while TOP ≠ 0:
CALL POP(STACK, ITEM)
Write: ITEM
[End of loop.]

4. Return.

- | | | |
|--|-----|--------|
| c) Consider the following arithmetic expression P, written in <i>postfix</i> notation: | 2.5 | CO3 C3 |
|--|-----|--------|

P: 20, 7, 3, -, /, 5, 2, 8, -, *, +

Evaluate P, using the algorithm you have studied.

- | | | |
|--|---|--------|
| d) Consider the following <i>infix</i> expression Q: | 3 | CO3 C3 |
|--|---|--------|

Q: A * (B - D) ↑ E + F * (G - H / K)

Translate Q into its equivalent *postfix* expression P using the algorithm you have studied.

au_22

3. a) Suppose the following numbers are stored in an array A:

22, 33, 29, 49, 21, 57, 62, 73, 54, 44

Apply *selection sort* algorithm to sort the array A and show each pass separately.

- b) Consider a situation where *swap* operation is very costly. Which of the following sorting algorithms should be preferred so that the number of swap operations are minimized in general? Why?

i) Heap Sort ii) Selection Sort iii) Insertion Sort iv) Merge Sort

- c) Write an algorithm to sort an array A of n elements using *insertion sort*.

- d) The following values are to be stored in a hash table:

25, 42, 96, 101, 102, 162, 197, 2XX

Describe how the values are hashed by using *division method* of hashing with a *table size* of 11. Use *linear probing* as the method of *collision resolution*.

[Here XX is the last two digits of your ID. For example, if ID is C191085, 2XX will be 285].

4. a) Consider the algebraic expression $E = (3m + n)(5x - y)^3$.

a) Draw the tree T which corresponds to the expression E.

b) Find the *preorder* and *postorder* of T.

- b) The *preorder* traversal sequence of a *binary search tree* is 30, 20, 10, 15, 25, 23, 39, 35, 1XX.

Draw the tree. Show each step.

[Here XX is the last two digits of your ID. For example, if ID is C191085, 1XX will be 185].

- c) What do you mean by *max heap*? Build a max heap from the following list of numbers:

30, 39, 29, 27, 37, 52, 63, 44

- d) Explain which data structure is most efficient to find the *top 10 largest items* out of *1 million items* stored in file? Why?

i) Min heap ii) Max heap iii) BST iv) Sorted array

OR

4. a) Suppose the following nine numbers are inserted in order into an empty binary search tree T:

50, 33, 44, 22, 77, 35, N, 60, 40,

i. Draw the tree T.

ii. Traverse the tree T in *preorder*, *postorder* and *inorder*.

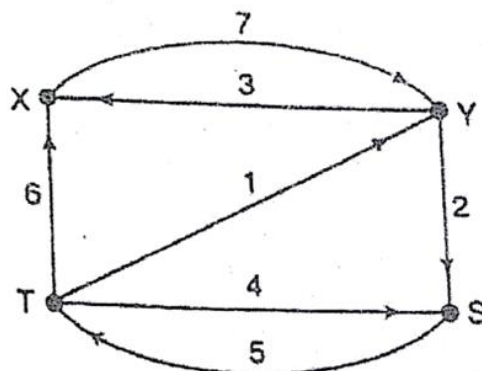
(Note: N is the last two digits of your ID. i.e. if your ID is C151216, then N=16)

- b) Illustrate the operation of Heap sort on the array $A = (5, 13, 2, 25, N, 7, 17, 20, 8, 4)$. (Note: N is the sum of last two digits of your ID. i.e. if your ID is C151216, then $N = 1 + 6 = 7$)

5. a) Describe the adjacency list and linked representation of graph with necessary figure.

- b) Consider the following weighted graph. Suppose the nodes are in an array DATA as follows:

Data: X, Y, S, T



Find the weighted matrix W of the weighted graph. Then find Q_0 and Q_1 from W using Warshall's algorithm.

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE

Final Examination, Spring 2022

Course Code: CSE 2321 Course Title: Data Structures
Time: 2 Hours 30 Minutes Total Marks: 50

[Answer all the following questions. Figures in the right hand margin indicate full marks. Use a Separate answer script for Group-A and Group-B.]

Group A

- | | | | |
|---|------|----|---|
| 1. a) Let N be an integer and suppose G(N) is recursively defined by | CLO | DL | |
| $G(N) = \begin{cases} 3 * N & \text{if } N < 5 \\ 2 * G(N - 5) + X & \text{otherwise} \end{cases}$ | CLO1 | C2 | 3 |
| i) Find the base criteria of G.
ii) Find G(2), G(8) and G(24).
[Here X is the last digit of your ID. For example, if ID is C191085, X will be 5]. | | | |
| b) Consider the following circular queue where Queue is allocated six memory cells: | CLO1 | C2 | 2 |
| FRONT=2, REAR=5 QUEUE: , , U, C, S, ,
Illustrate the Queue, including FRONT and REAR , as the following sequences take place:
i) E is added iii) U, V, A are added
ii) Two items are deleted iv) Three items are deleted | | | |
| c) Write a procedure to insert an element into a circular queue and delete an element from a circular queue. | CLO3 | C4 | 5 |

Or

Tower of Hanoi consists of three pegs or towers with n disks placed one over the other. The objective of the puzzle is to move the stack to another peg following these simple rules. Only one disk can be moved at a time. No disk can be placed on top of the smaller disk.
 How many moves does it take to solve the Tower of Hanoi for 5 disks?

- | | | | |
|--|---|-----|--|
| 3. a) Suppose the following stack of integers is in memory where STACK is allocated N = 8 memory cells:
TOP = 4 STACK: 11, 22 N, 33, , , , ,
Describe the stack as the following operations take place –
i) POP (STACK, ITEM) iv) PUSH (STACK, 66)
ii) PUSH (STACK, 44) v) POP (STACK, ITEM)
iii) PUSH (STACK, 55) vi) POP (STACK, ITEM)
[Here N is the last two digits of our ID, i.e. if your ID is C193216, then N=16] | 1 | CO1 | |
| b) Suppose stacks A[1] and A[2] are stored in a linear array STACK with N elements, as pictured in Fig. 6.37. Assume TOP[K] denotes the top of stack A[K].
(a) Write a procedure PUSH(STACK, N, TOP, ITEM, K) which pushes ITEM onto stack A[K].
(b) Write a procedure POP(STACK, TOP, ITEM, K) which deletes the top element from stack A[K] and assigns the element to the variable ITEM. | 2 | CO4 | |

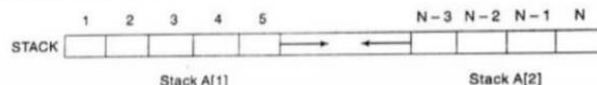


Fig. 6.37

- | | | | |
|--|---|-----|--|
| c) Evaluate each of the following postfix expression using the algorithm that you have studied.
P₁: N, 7, 3, +, *, 5, 2, 8, -, /, +
P₂: 3, 1, +, 2, /, N, 4, -, 2, /, +, 5, *
[Here, N is the last digit of your ID. For example, if ID = C193009 then N = 9] | 2 | CO3 | |
| d) Consider the following infix expression Q:
Q: (A * (B - D) / E + F) * (G - H / K)
Translate Q into its equivalent postfix expression P using the algorithm you studied. | 2 | CO3 | |

sp_22

Group B

3. a) Illustrate the hash function. CLO3 C2 2
 b) Research the following algorithms to compare their complexity: CLO2 C3 4
 i) Linear Search Algorithm
 ii) Bubble Sort Algorithm
 iii) Binary Search Algorithm
 iv) Insertion Sort Algorithm
 v) Selection Sort Algorithm
 c) Suppose the following characters are stored in an array A: CLO3 C2 4
 B, A, N, G, L, A, D, E, S, H
 Apply *selection sort* algorithm to sort the array A and show each pass separately.
 Or

Suppose the following characters are stored in an array A: CLO3 C2 4
 D, A, T, A, S, T, R, U, C, T, U, R, E, S
 Apply *insertion sort* algorithm to sort the array A and show each pass separately.

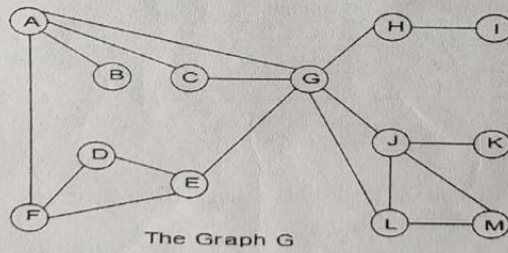
4. a) Analyze the differences between Binary tree and Complete binary tree with figure. CLO1 C2 5
 Construct an *expression tree* T for the following arithmetic expression:
 $(A + B * C) + ((D / E - F) * G)$
 Traverse the tree T in **preorder**, **postorder** and **inorder**.
 b) Explain which data structure is most efficient to find the *top 10 largest items* out of 1 million items stored in file? Why? CLO4 C4 1
 i) Min heap ii) Max heap iii) BST iv) Sorted array
 c) What is a *binary search tree*? Suppose the following eight numbers are inserted in order into an empty binary search tree. CLO1 C2 4
 49, 42, 19, 62, 37, 79, 44, 8
 i) Draw the tree T.
 ii) Find the inorder traversal of T.

Or

What do you mean by *max heap*? Build a max heap from the following list of numbers CLO1 C2 4
 (Show each step):
 33, 29, 49, 21, 57, 62, 73, 54

5. a) Define *graph*? Draw a picture of the directed graph specified below: CLO1 C2 3
 $G = (V, E)$
 $V(G) = \{1, 2, 3, 4, 5, 6\}$
 $E(G) = \{(1, 2), (2, 3), (3, 4), (5, 1), (5, 6), (2, 6), (1, 6), (4, 6), (2, 4)\}$
 Obtain the following for the above graph:
 i) Find the *adjacency matrix* A of the graph G.
 ii) Find the *adjacency list* of the graph G.
 b) For a set of vertexes V with n elements, how many possible edges there? CLO2 C2 2
 c) Give a comparison between *breadth first search* and *depth first search*. CLO1 C4 1

- d) Traverse the graph G shown below in **breadth first order**, **depth first order** and construct the **breadth first** and **depth first spanning trees**. Start from node C [if last digit of your ID is *odd*] / J [if last digit of your ID is *even*].



Or

Analyze the following graph G to

- i. Evaluate the adjacency matrix A of Graph G.
- ii. Evaluate the path matrix P of Graph G.

